

4. Project Planning Phase

1. INTRODUCTION

The Project Planning Phase is the fourth stage in the development of **LegalEase AI**. After completing the requirement analysis and system design, the project activities are organized into a systematic development plan.

In this phase, the required software, libraries, APIs, development tools, and project tasks are identified. The development process is divided into different stages such as frontend development, backend development, AI integration, testing, and documentation.

The main purpose of this phase is to make the development process organized and systematic while considering possible technical and integration challenges.

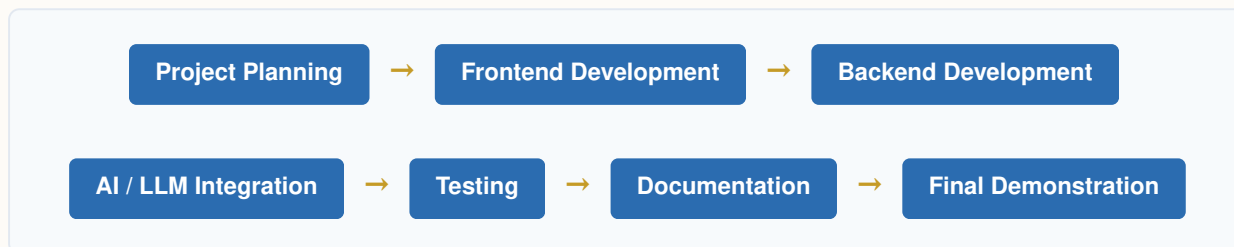
2. OBJECTIVES

The main objectives of the Project Planning Phase are:

- Divide the project into different development stages.
- Identify the required software and development tools.
- Identify required libraries and APIs.
- Organize tasks based on priority and dependencies.
- Plan frontend and backend development.
- Plan AI/LLM integration.
- Plan testing and documentation activities.
- Identify possible technical and integration challenges.

3. PROJECT DEVELOPMENT PLAN

The development of LegalEase AI is divided into structured, sequential stages. Each stage is planned according to its operational requirements and technical dependencies:



4. TASK PLANNING

The project documentation specifically identifies frontend, backend, AI integration, testing, and documentation as activities to be planned:

Task	Main Activity
Frontend	Create user interface
Backend	Develop APIs and request handling
AI Integration	Connect AI/LLM service
Configuration	Configure API keys and dependencies
Testing	Test inputs, outputs, and API connectivity
Documentation	Record development and testing details

5. TOOLS AND TECHNOLOGIES

The planning phase identifies the resources required for development before implementation begins so that the development process can proceed systematically:

- **VS Code:** Development environment for authoring and debugging code.
- **Python:** Core programming language for backend operations.
- **Flask:** Lightweight framework for backend and API development.
- **AI / LLM API:** Service providing intelligent natural language processing.
- **Python Libraries:** Application dependencies for data processing and API interaction.
- **Git / GitHub:** Version control management and collaborative repository hosting.
- **.env Configuration:** Environment file for secure API key and secrets storage.

6. BASIC PROJECT STRUCTURE

A planned project structure is organized to maintain modularity and maintainability:

```
LegalEase-AI/  
|  
├─ app/  
│  ├── __init__.py  
│  └── routes.py  
|  
├─ templates/  
│  └── index.html  
|  
├─ static/  
│  └── style.css  
|
```

```
|  
|— requirements.txt  
|— .env  
|— README.md
```

Structure Explanation:

- `app/` – Contains the main backend application logic.
- `routes.py` – Contains API routes for receiving and processing user requests.
- `templates/` – Contains the frontend HTML files.
- `static/` – Contains CSS and other static frontend resources.
- `requirements.txt` – Lists the Python dependencies required for the project.
- `.env` – Stores configuration values such as environment variables and API keys.
- `README.md` – Contains project overview, documentation, and setup instructions.

7. RISK AND CHALLENGE PLANNING

Possible technical challenges are evaluated during the planning stage. Anticipating these risks helps the development team prepare appropriate fallback solutions prior to implementation:

- AI/LLM API connectivity disruptions or rate limits.
- Incorrect or missing API key configurations.
- Dependency installation and version compatibility errors.
- Backend and frontend integration discrepancies.
- Unexpected or non-deterministic AI-generated responses.
- Testing edge cases across varied legal domain user queries.

8. EXPECTED OUTCOME

At the conclusion of the Project Planning Phase, the development workflow of LegalEase AI is clearly established. Expected deliverables include defined stages, selected software and tools, dependency mappings, task priorities, planned architecture, testing procedures, and risk mitigation strategies.

9. PHASE CONCLUSION

The Project Planning Phase provides a structured roadmap for the development of LegalEase AI. It organizes the essential activities required to convert system designs into a functional application. By establishing clear task dependencies and technical guidelines, the project seamlessly transitions into the **Project Development Phase** for implementation.